# 0707.4470: equation evidence

## Display equations

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<tr><td>0707.4470_EQ0025</td><td>17</td><td>0.382</td><td><code>\begin{aligned} & F=E_{x} \mathrm{~d} x \wedge \mathrm{~d} t+E_{y} \mathrm{~d} y \wedge \mathrm{~d} t+E_{z} \mathrm{~d} z \wedge d t \\ & \quad+B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y \end{aligned}</code></td><td>$$\begin{aligned} & F=E_{x} \mathrm{~d} x \wedge \mathrm{~d} t+E_{y} \mathrm{~d} y \wedge \mathrm{~d} t+E_{z} \mathrm{~d} z \wedge d t \\ & \quad+B_{x} \mathrm{~d} y \wedge \mathrm{~d} z+B_{y} \mathrm{~d} z \wedge \mathrm{~d} x+B_{z} \mathrm{~d} x \wedge \mathrm{~d} y \end{aligned}$$</td><td>$$\begin{aligned} & F=E_{x} \mathrm{d} x \wedge \mathrm{d} t+E_{y} \mathrm{d} y \wedge \mathrm{d} t+E_{z} \mathrm{d} z \wedge d t \\ & \quad+B_{x} \mathrm{d} y \wedge \mathrm{d} z+B_{y} \mathrm{d} z \wedge \mathrm{d} x+B_{z} \mathrm{d} x \wedge \mathrm{d} y, \end{aligned}$$</td></tr>
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<tr><td>0707.4470_EQ0026</td><td>18</td><td>■ 0.856</td><td><code>\begin{gathered} x t \text {-face : }\left.E_{x} \right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}} \Delta x \Delta t \\ y t \text {-face : }\left.E_{y}\right|_{k, l+ \frac{1}{2}, m} ^{n+\frac{1}{2}} \Delta y \Delta t \\ z t \text {-face : }\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}} \Delta z \Delta t \\ y z \text {-face : }\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n} \Delta y \Delta z \\ x z \text {-face : }\left.B_{y} \right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n} \Delta z \Delta x \\ x y \text {-face : }\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m} ^{n} \Delta x \Delta y \end{gathered}</code></td><td>$$\begin{gathered} xt\text{-face : }\left.E_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ yt\text{-face : }\left.E_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ zt\text{-face : }\left.E_{z}\right|_{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \Delta z \Delta t \\ yz\text{-face : }\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}}^{n} \Delta y \Delta z \\ xz\text{-face : }\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}}^{n} \Delta z \Delta x \\ xy\text{-face : }\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n} \Delta x \Delta y \end{gathered}$$</td><td>$$\begin{gathered} xt\text{-face}: \left.E_{x}\right|_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ yt\text{-face}: \left.E_{y}\right|_{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ zt\text{-face}: \left.E_{z}\right|_{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} \Delta z \Delta t \\ yz\text{-face}: \left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}}^{n} \Delta y \Delta z \\ xz\text{-face}: \left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}}^{n} \Delta z \Delta x \\ xy\text{-face}: \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n} \Delta x \Delta y. \end{gathered}$$</td></tr>
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<tr><td>0707.4470_EQ0030</td><td>18</td><td>0.881</td><td><code>\begin{aligned} y z t \text {-face: } & -\left( \left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left( \left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+ \frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}</code></td><td>$\begin{aligned} y z t \text {-face: } & -\left( \left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left( \left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+ \frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}$</td><td>$\begin{aligned} y z t \text {-face: } & -\left( \left.E_{y}\right|_{k, l+\frac{1}{2}, m+1} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & + \left(\left.E_{z}\right|_{k, l+1, m+\frac{1}{2}} ^{n+ \frac{1}{2}}-\left.E_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}\right) \Delta z \Delta t \\ & +\left( \left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n+1}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+ \frac{1}{2}} ^{n}\right) \Delta y \Delta z \end{aligned}$</td></tr>
<tr><td>0707.4470_ EQ0058 (5.3)</td><td>32</td><td>0.887</td><td><code>\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} \left(\mathrm{~d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{~d} t+\alpha \wedge J \wedge \mathrm{~d} t\right) \\ & =\int_{X} \alpha \wedge\left(\mathrm{~d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{~d} t+J \wedge \mathrm{~d} t\right) \end{aligned}</code></td><td>$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} \left(\mathrm{~d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{~d} t+\alpha \wedge J \wedge \mathrm{~d} t\right) \\ & =\int_{X} \alpha \wedge\left(\mathrm{~d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{~d} t+J \wedge \mathrm{~d} t\right) \end{aligned}$</td><td>$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\int_{X} \left(\mathrm{d}_{t} \alpha \wedge D-\mathrm{d}_{s} \alpha \wedge H \wedge \mathrm{d} t+\alpha \wedge J \wedge \mathrm{d} t\right) \\ & =\int_{X} \alpha \wedge\left(\mathrm{d}_{t} D-\mathrm{d}_{s} H \wedge \mathrm{d} t+J \wedge \mathrm{d} t\right). \end{aligned}$</td></tr>
<tr><td>0707.4470_EQ0059</td><td>32</td><td>0.892</td><td><code>\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},</code></td><td>$\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},$</td><td>$\mathbf{d} S[A] \cdot \alpha=\left.\int_{\Sigma} \alpha \wedge D\right|_{t_{0}} ^{t_{f}},$</td></tr>
<tr><td>0707.4470_ EQ0016 (2.5)</td><td>8</td><td>0.911</td><td><code>\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F- \mathrm{d} A) \wedge \gamma] \\ & =\int_{X}[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F- \mathrm{d} A) \wedge \gamma] \end{aligned}</code></td><td>$\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F- \mathrm{d} A) \wedge \gamma] \\ & =\int_{X}[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F- \mathrm{d} A) \wedge \gamma] \end{aligned}$</td><td>$\begin{aligned} \mathbf{d} S[A, F, G] \cdot(\alpha, \phi, \gamma) & =\int_{X}[-\phi \wedge * F+\alpha \wedge \mathscr{J}+(\phi-\mathrm{d} \alpha) \wedge G+(F- \mathrm{d} A) \wedge \gamma] \\ & =\int_{X}[\alpha \wedge(\mathscr{J}-\mathrm{d} G)+\phi \wedge(G-* F)+(F- \mathrm{d} A) \wedge \gamma]\,. \end{aligned}$</td></tr>
<tr><td>0707.4470_EQ0028</td><td>18</td><td>0.937</td><td><code>\begin{aligned} x y t \text {-face : } & -\left( \left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left( \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}</code></td><td>$\begin{aligned} x y t \text {-face : } & -\left( \left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left( \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}$</td><td>$\begin{aligned} x y t \text {-face : } & -\left( \left.E_{x}\right|_{k+\frac{1}{2}, l+1, m} ^{n+ \frac{1}{2}}-\left.E_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}\right) \Delta x \Delta t \\ & + \left(\left.E_{y}\right|_{k+1, l+\frac{1}{2}, m} ^{n+ \frac{1}{2}}-\left.E_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}\right) \Delta y \Delta t \\ & +\left( \left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n+1}-\left.B_{z}\right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}\right) \Delta x \Delta y \end{aligned}$</td></tr>
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<tr><td>0707.4470_ EQ0036 (4.2)</td><td>20</td><td>0.951</td><td><code>\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x}\right|_{k-\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l-\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+\frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m} ^{n+\frac{1}{2}} . \end{aligned}</code></td><td>$\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x}\right|_{k-\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l-\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+\frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m} ^{n+\frac{1}{2}} . \end{aligned}$</td><td>$\begin{aligned} & \frac{\left.D_{x}\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}-\left.D_{x}\right|_{k-\frac{1}{2}, l, m} ^{n+\frac{1}{2}}}{\Delta x}+\frac{\left.D_{y}\right|_{k, l+\frac{1}{2}, m} ^{n+\frac{1}{2}}-\left.D_{y}\right|_{k, l-\frac{1}{2}, m} ^{n+\frac{1}{2}}}{\Delta y} \\ & \quad+\frac{\left.D_{z}\right|_{k, l, m+\frac{1}{2}} ^{n+\frac{1}{2}}-\left.D_{z}\right|_{k, l, m-\frac{1}{2}} ^{n+\frac{1}{2}}}{\Delta z}=\left.\rho\right|_{k, l, m} ^{n+\frac{1}{2}} . \end{aligned}$</td></tr>
<tr><td>0707.4470_EQ0023</td><td>13</td><td>0.955</td><td><code>\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle \langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}</code></td><td>$\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle \langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}$</td><td>$\begin{aligned} (\alpha, \beta) & =\sum_{\sigma^{k}} \kappa(\sigma)\binom{n}{k} \frac{|\mathrm{CH}(\sigma, * \sigma)|}{|\sigma|^{2}}\langle\alpha, \sigma\rangle \langle\beta, \sigma\rangle \\ & =\sum_{\sigma^{k}} \kappa(\sigma) \frac{|* \sigma|}{|\sigma|}\langle\alpha, \sigma\rangle\langle\beta, \sigma\rangle \end{aligned}$</td></tr>
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<tr><td>0707.4470_ EQ0012 (2.3)</td><td>8</td><td>0.982</td><td><code>\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{~d} G=\mathscr{J} \end{aligned}</code></td><td>$\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{~d} G=\mathscr{J} \end{aligned}$</td><td>$\begin{aligned} & \mathrm{d} F=0 \\ & \mathrm{d} G=\mathscr{J} \end{aligned}$</td></tr>
<tr><td>0707.4470_EQ0060</td><td>32</td><td>0.982</td><td><code>\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{\Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}} ^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{~d}_{s} D \right|_{t_{0}} ^{t_{f}}</code></td><td>$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{\Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}} ^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{~d}_{s} D \right|_{t_{0}} ^{t_{f}}$</td><td>$\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\left.\int_{\Sigma} \mathrm{d}_{s} f \wedge D\right|_{t_{0}} ^{t_{f}}=-\left.\int_{\Sigma} f \wedge \mathrm{d}_{s} D \right|_{t_{0}} ^{t_{f}}$</td></tr>
<tr><td>0707.4470_ EQ0046 (4.4)</td><td>24</td><td>0.988</td><td><code>\frac{D_{e}^{m+1 / 2}-D_{e}^{m-1 / 2}}{t_{e}^{m+1 / 2}-t_{e}^{m-1 / 2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}\right)-J_{e}^{m},</code></td><td>$\frac{D_{e}^{m+1 / 2}-D_{e}^{m-1 / 2}}{t_{e}^{m+1 / 2}-t_{e}^{m-1 / 2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}\right)-J_{e}^{m},$</td><td>$\frac{D_{e}^{m+1/2}-D_{e}^{m-1/2}}{t_{e}^{m+1/2}-t_{e}^{m-1/2}}=\mathrm{d}_{1}^{T}\left(H_{\sigma}^{n} \nVdash_{\{t_{\sigma}^{n}=t_{e}^{m}\}}\right)-J_{e}^{m},$</td></tr>
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<tr><td>0707.4470_ EQ0013 (2.4)</td><td>8</td><td>0.996</td><td><code>\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}</code></td><td>$\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}$</td><td>$\begin{aligned} \nabla \times \mathbf{E}+\partial_{t} \mathbf{B} & =0 \\ \nabla \cdot \mathbf{B} & =0 . \end{aligned}$</td></tr>
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<tr><td>0707.4470_EQ0015 (2.7)</td><td>8</td><td>0.999</td><td><code>S[A, F, G]=\int_{X}\left[-\frac{1}{2} F \wedge * F+A \wedge \mathscr{J}+(F-\mathrm{d} A) \wedge G\right] .</code></td><td>$S[A, F, G]=\int_{X}\left[-\frac{1}{2} F \wedge * F+A \wedge \mathscr{J}+(F-\mathrm{d} A) \wedge G\right] .$</td><td>$S[A,F,G]=\int_X\left[-\frac{1}{2}F\wedge*F+A\wedge\mathscr{J}+(F-\mathrm{d}A)\wedge G\right].$</td></tr>
<tr><td>0707.4470_EQ0010</td><td>7</td><td>1.000</td><td><code>\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\left. \frac{\mathrm{d}}{\mathrm{~d} \epsilon}\right|_{ \epsilon=0} S[A+\epsilon \alpha] \\ & =\int_{X}(- \mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}) \\ & =\int_{X} \alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), \end{aligned}</code></td><td>$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & =\left. \frac{\mathrm{d}}{\mathrm{~d} \epsilon}\right|_{ \epsilon=0} S[A+\epsilon \alpha] \\ & =\int_{X}(- \mathrm{d} \alpha \wedge * \mathrm{~d} A+\alpha \wedge \mathscr{J}) \\ & =\int_{X} \alpha \wedge(-\mathrm{d} * \mathrm{~d} A+\mathscr{J}), \end{aligned}$</td><td>$\begin{aligned} \mathbf{d}S[A]\cdot\alpha & =\left.\frac{\mathrm{d}}{\mathrm{d}\epsilon}\right|_{\epsilon=0}S[A+\epsilon\alpha] \\ & =\int_X(-\mathrm{d}\alpha\wedge*\mathrm{d}A+\alpha\wedge\mathscr{J}) \\ & =\int_X\alpha\wedge(-\mathrm{d}*\mathrm{d}A+\mathscr{J}), \end{aligned}$</td></tr>
<tr><td>0707.4470_EQ0033 (4.1)</td><td>19</td><td>1.000</td><td><code>\begin{aligned} & \frac{\left.B_{x}\right|_{k+1, l+ \frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{ \left.B_{y}\right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0 . \end{aligned}</code></td><td>$\begin{aligned} & \frac{\left.B_{x}\right|_{k+1, l+ \frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{ \left.B_{y}\right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0 . \end{aligned}$</td><td>$\begin{aligned} & \frac{\left.B_{x}\right|_{k+1, l+ \frac{1}{2}, m+\frac{1}{2}} ^{n}-\left.B_{x}\right|_{k, l+\frac{1}{2}, m+\frac{1}{2}} ^{n}}{\Delta x}+\frac{ \left.B_{y}\right|_{k+\frac{1}{2}, l+1, m+\frac{1}{2}} ^{n}-\left.B_{y}\right|_{k+\frac{1}{2}, l, m+\frac{1}{2}} ^{n}}{\Delta y} \\ & \quad+\frac{\left.B_{z}\right|_{k+ \frac{1}{2}, l+\frac{1}{2}, m+1} ^{n}-\left.B_{z} \right|_{k+\frac{1}{2}, l+\frac{1}{2}, m} ^{n}}{\Delta z}=0. \end{aligned}$</td></tr>
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<tr><td>0707.4470_EQ0003</td><td>6</td><td>1.000</td><td><code>\oint_{C} \mathbf{H} \cdot \mathrm{~d} \mathbf{l}=\frac{ \mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{D} \cdot \mathrm{~d} \mathbf{A},</code></td><td>$\oint_{C} \mathbf{H} \cdot \mathrm{~d} \mathbf{l}=\frac{ \mathrm{d}}{\mathrm{~d} t} \int_{S} \mathbf{D} \cdot \mathrm{~d} \mathbf{A},$</td><td>$\oint_C\mathbf{H}\cdot\mathrm{d}\mathbf{l}=\frac{\mathrm{d}}{\mathrm{d}t}\int_S\mathbf{D}\cdot\mathrm{d}\mathbf{A},$</td></tr>
<tr><td>0707.4470_EQ0004</td><td>6</td><td>1.000</td><td><code>\mathbf{D}=\epsilon \mathbf{E}, \quad \mathbf{B}=\mu \mathbf{H} .</code></td><td>$\mathbf{D}=\epsilon \mathbf{E}, \quad \mathbf{B}=\mu \mathbf{H} .$</td><td>$\mathbf{D}=\epsilon\mathbf{E},\qquad\mathbf{B}=\mu\mathbf{H}.$</td></tr>
<tr><td>0707.4470_EQ0005</td><td>6</td><td>1.000</td><td><code>F=E \wedge \mathrm{~d} t+B .</code></td><td>$F=E \wedge \mathrm{~d} t+B .$</td><td>$F=E\wedge\mathrm{d}t+B.$</td></tr>
<tr><td>0707.4470_EQ0006</td><td>7</td><td>1.000</td><td><code>G=* F=H \wedge \mathrm{~d} t-D,</code></td><td>$G=* F=H \wedge \mathrm{~d} t-D,$</td><td>$G=*F=H\wedge\mathrm{d}t-D,$</td></tr>
<tr><td>0707.4470_EQ0007</td><td>7</td><td>1.000</td><td><code>\mathscr{J}=J \wedge \mathrm{~d} t-\rho .</code></td><td>$\mathscr{J}=J \wedge \mathrm{~d} t-\rho .$</td><td>$\mathscr{J}=J\wedge\mathrm{d}t-\rho.$</td></tr>
</table>

| Identifier | Page | Conf. | LaTeX source | Rendered | Image |
|---|---|---|---|---|---|
| 0707.4470_ EQ0011 (2.1) | 7 | 1.000 | `\mathrm{d} * \mathrm{~d} A=\mathscr{J} .` | $\mathrm{d} * \mathrm{~d} A=\mathscr{J} .$ | $\mathrm{d} * \mathrm{d} A=\mathscr{J} .$ |
| 0707.4470_ EQ0014 (2.6) | 8 | 1.000 | `\begin{aligned} \nabla \times \mathbf{H}-\partial_{t} \mathbf{D} & =\mathbf{J} \\ \nabla \cdot \mathbf{D} & = \rho . \end{aligned}` | $\begin{aligned} \nabla \times \mathbf{H}-\partial_{t} \mathbf{D} & =\mathbf{J} \\ \nabla \cdot \mathbf{D} & = \rho . \end{aligned}$ | $\begin{aligned} \nabla \times \mathbf{H}-\partial_{t} \mathbf{D} & =\mathbf{J} \\ \nabla \cdot \mathbf{D} & = \rho . \end{aligned}$ |
| 0707.4470_ EQ0017 (2.2) | 8 | 1.000 | `\mathrm{d} G=\mathscr{J}, \quad G=* F, \quad F=\mathrm{d} A .` | $\mathrm{d} G=\mathscr{J}, \quad G=* F, \quad F=\mathrm{d} A .$ | $\mathrm{d} G=\mathscr{J}, \quad G=* F, \quad F=\mathrm{d} A .$ |
| 0707.4470_EQ0018 | 9 | 1.000 | `\partial_{t}(\nabla \cdot \mathbf{B})=0, \quad \partial_{t}(\nabla \cdot \mathbf{D})+\nabla \cdot \mathbf{J}=\partial_{t}(\nabla \cdot \mathbf{D}-\rho)=0 .` | $\partial_{t}(\nabla \cdot \mathbf{B})=0, \quad \partial_{t}(\nabla \cdot \mathbf{D})+\nabla \cdot \mathbf{J}=\partial_{t}(\nabla \cdot \mathbf{D}-\rho)=0 .$ | $\partial_{t}(\nabla \cdot \mathbf{B})=0, \quad \partial_{t}(\nabla \cdot \mathbf{D})+\nabla \cdot \mathbf{J}=\partial_{t}(\nabla \cdot \mathbf{D}-\rho)=0 .$ |
| 0707.4470_EQ0019 | 12 | 1.000 | `\langle\alpha, \sigma\rangle \equiv \int_{\sigma} \alpha .` | $\langle\alpha, \sigma\rangle \equiv \int_{\sigma} \alpha .$ | $\langle\alpha, \sigma\rangle \equiv \int_{\sigma} \alpha .$ |
| 0707.4470_EQ0021 | 12 | 1.000 | `\langle\mathrm{d} \alpha, \sigma\rangle=\langle\alpha, \partial \sigma\rangle,` | $\langle\mathrm{d} \alpha, \sigma\rangle=\langle\alpha, \partial \sigma\rangle,$ | $\langle\mathrm{d} \alpha, \sigma\rangle=\langle\alpha, \partial \sigma\rangle,$ |
| 0707.4470_EQ0022 | 13 | 1.000 | `\frac{1}{\|* \sigma\|}\langle * \alpha, * \sigma\rangle= \kappa(\sigma) \frac{1}{\|\sigma\|}\langle\alpha, \sigma \rangle,` | $\frac{1}{\lvert * \sigma\rvert}\langle * \alpha, * \sigma\rangle= \kappa(\sigma) \frac{1}{\lvert\sigma\rvert}\langle\alpha, \sigma \rangle,$ | $\frac{1}{\lvert * \sigma\rvert}\langle * \alpha, * \sigma\rangle= \kappa(\sigma) \frac{1}{\lvert\sigma\rvert}\langle\alpha, \sigma \rangle,$ |
| 0707.4470_ EQ0024 (3.1) | 16 | 1.000 | `(\mathrm{d} \alpha, \beta)=(\alpha, \delta \beta)+\langle \alpha \wedge * \beta, \partial K\rangle .` | $(\mathrm{d} \alpha, \beta)=(\alpha, \delta \beta)+\langle \alpha \wedge * \beta, \partial K\rangle .$ | $(\mathrm{d} \alpha, \beta)=(\alpha, \delta \beta)+\langle \alpha \wedge * \beta, \partial K\rangle .$ |
| 0707.4470_EQ0027 | 18 | 1.000 | `\mathrm{d} F=0, \quad \mathrm{~d} G=\mathscr{J},` | $\mathrm{d} F=0, \quad \mathrm{~d} G=\mathscr{J},$ | $\mathrm{d} F=0, \quad \mathrm{d} G=\mathscr{J},$ |
| 0707.4470_EQ0034 | 19 | 1.000 | `\partial_{t} \mathbf{B}=-\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B}=0 .` | $\partial_{t} \mathbf{B}=-\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B}=0 .$ | $\partial_{t} \mathbf{B}=-\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B}=0 .$ |
| 0707.4470_EQ0037 | 20 | 1.000 | `\partial_{t} \mathbf{D}=\nabla \times \mathbf{H}- \mathbf{J}, \quad \nabla \cdot \mathbf{D}=\rho .` | $\partial_{t} \mathbf{D}=\nabla \times \mathbf{H}- \mathbf{J}, \quad \nabla \cdot \mathbf{D}=\rho .$ | $\partial_{t} \mathbf{D}=\nabla \times \mathbf{H}- \mathbf{J}, \quad \nabla \cdot \mathbf{D}=\rho .$ |
| 0707.4470_EQ0038 | 21 | 1.000 | `F=E \wedge \mathrm{~d} t+B .` | $F=E \wedge \mathrm{~d} t+B .$ | $F=E \wedge \mathrm{d} t+B .$ |
| 0707.4470_EQ0039 | 22 | 1.000 | `\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1 / 2} .` | $\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1 / 2} .$ | $\frac{B^{n+1}-B^{n}}{\Delta t}=-\mathrm{d}_{1} E^{n+1 / 2} .$ |
| 0707.4470_EQ0040 | 22 | 1.000 | `\frac{D^{n+1 / 2}-D^{n-1 / 2}}{\Delta t}= \mathrm{d}_{1}^{T} H^{n}-J^{n} .` | $\frac{D^{n+1 / 2}-D^{n-1 / 2}}{\Delta t}= \mathrm{d}_{1}^{T} H^{n}-J^{n} .$ | $\frac{D^{n+1 / 2}-D^{n-1 / 2}}{\Delta t}= \mathrm{d}_{1}^{T} H^{n}-J^{n} .$ |
| 0707.4470_EQ0041 | 23 | 1.000 | `\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{ \sigma}^{N_{\sigma}}\right\} .` | $\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{ \sigma}^{N_{\sigma}}\right\} .$ | $\Theta_{\sigma}=\left\{t_{\sigma}^{0}<\cdots<t_{ \sigma}^{N_{\sigma}}\right\} .$ |
| 0707.4470_EQ0042 | 23 | 1.000 | `\Theta_{e}=\bigcup_{\sigma \ni e} \Theta_{\sigma}=\left \{t_{e}^{0} \leq \cdots \leq t_{e}^{N_{e}}\right\} .` | $\Theta_{e}=\bigcup_{\sigma \ni e} \Theta_{\sigma}=\left \{t_{e}^{0} \leq \cdots \leq t_{e}^{N_{e}}\right\} .$ | $\Theta_{e}=\bigcup_{\sigma \ni e} \Theta_{\sigma}=\left \{t_{e}^{0} \leq \cdots \leq t_{e}^{N_{e}}\right\} .$ |
| 0707.4470_EQ0043 | 23 | 1.000 | `\Theta_{e}^{\prime}=\left\{t_{e}^{1 / 2} \leq \cdots \leq t_{e}^{N_{e}-1 / 2}\right\},` | $\Theta_{e}^{\prime}=\left\{t_{e}^{1 / 2} \leq \cdots \leq t_{e}^{N_{e}-1 / 2}\right\},$ | $\Theta_{e}^{\prime}=\left\{t_{e}^{1 / 2} \leq \cdots \leq t_{e}^{N_{e}-1 / 2}\right\},$ |
| 0707.4470_EQ0044 | 24 | 1.000 | `\Theta_{* \sigma}=\Theta_{\sigma}, \quad \Theta_{* e}= \Theta_{e} .` | $\Theta_{* \sigma}=\Theta_{\sigma}, \quad \Theta_{* e}= \Theta_{e} .$ | $\Theta_{* \sigma}=\Theta_{\sigma}, \quad \Theta_{* e}= \Theta_{e} .$ |
| 0707.4470_ EQ0045 (4.3) | 24 | 1.000 | `\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{ \sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1} \sum\left \{E_{e}^{m+1 / 2}: t_{\sigma}^{n}<t_{e}^{m+1 / 2}<t_{ \sigma}^{n+1}\right\} .` | $\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{ \sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1} \sum\left \{E_{e}^{m+1 / 2}: t_{\sigma}^{n}<t_{e}^{m+1 / 2}<t_{ \sigma}^{n+1}\right\} .$ | $\frac{B_{\sigma}^{n+1}-B_{\sigma}^{n}}{t_{ \sigma}^{n+1}-t_{\sigma}^{n}}=-\mathrm{d}_{1} \sum\left \{E_{e}^{m+1 / 2}: t_{\sigma}^{n}<t_{e}^{m+1 / 2}<t_{ \sigma}^{n+1}\right\} .$ |
| 0707.4470_EQ0048 | 29 | 1.000 | `S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle .` | $S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle .$ | $S_{d}[A]=\left\langle\mathscr{L}_{d}, K\right\rangle .$ |
| 0707.4470_ EQ0051 (5.1) | 30 | 1.000 | `\mathbf{d} S_{d}(A) \cdot \alpha=\langle\alpha \wedge * \mathrm{~d} A, \partial K\rangle` | $\mathbf{d} S_{d}(A) \cdot \alpha=\langle\alpha \wedge * \mathrm{~d} A, \partial K\rangle$ | $\mathbf{d} S_{d}(A) \cdot \alpha=\langle\alpha \wedge * \mathrm{d} A, \partial K\rangle$ |

| Identifier | Page | Conf. | LaTeX source | Rendered | Image |
|---|---|---|---|---|---|
| 0707.4470_EQ0052 | 30 | ■ 1.000 | `\theta_{\mathscr{L}_{d}} \cdot \alpha=\alpha \wedge * \mathrm{~d} A .` | $\theta_{\mathscr{L}_{d}} \cdot \alpha=\alpha \wedge * \mathrm{~d} A .$ | $\theta_{\mathscr{L}_d} \cdot \alpha = \alpha \wedge * \mathrm{d}A.$ |
| 0707.4470_EQ0053 | 30 | ■ 1.000 | `\mathbf{d}^{2} S_{d}[A] \cdot \alpha \cdot \beta=\langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle .` | $\mathbf{d}^{2} S_{d}[A] \cdot \alpha \cdot \beta=\langle \mathbf{d} \theta \cdot \alpha \cdot \beta, \partial K \rangle .$ | $\mathbf{d}^2 S_d[A] \cdot \alpha \cdot \beta = \langle \mathbf{d}\theta \cdot \alpha \cdot \beta, \partial K \rangle .$ |
| 0707.4470_ EQ0054 (5.2) | 30 | ■ 1.000 | `\left\langle\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta, \partial K\right\rangle=0` | $\left\langle\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta, \partial K\right\rangle=0$ | $\langle \omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta, \partial K \rangle = 0$ |
| 0707.4470_EQ0055 | 31 | ■ 1.000 | `A=A_{x} \mathrm{~d} x+A_{y} \mathrm{~d} y+A_{z} \mathrm{~d} z .` | $A=A_{x} \mathrm{~d} x+A_{y} \mathrm{~d} y+A_{z} \mathrm{~d} z .$ | $A = A_x \,\mathrm{d}x + A_y \,\mathrm{d}y + A_z \,\mathrm{d}z.$ |
| 0707.4470_EQ0056 | 32 | ■ 1.000 | `\mathrm{d}_{t} A=E \wedge \mathrm{~d} t, \quad \mathrm{~d}_{s} A=B .` | $\mathrm{d}_{t} A=E \wedge \mathrm{~d} t, \quad \mathrm{~d}_{s} A=B .$ | $\mathrm{d}_t A = E \wedge \mathrm{d}t, \qquad \mathrm{d}_s A = B.$ |
| 0707.4470_EQ0057 | 32 | ■ 1.000 | `\begin{aligned} \mathscr{L} & =-\frac{1}{2}\left( \mathrm{~d}_{t} A+\mathrm{d}_{s} A\right) \wedge *\left( \mathrm{~d}_{t} A+\mathrm{d}_{s} A\right)+A \wedge \mathscr{J} \\ & =-\frac{1}{2}\left(\mathrm{~d}_{t} A \wedge * \mathrm{~d}_{t} A+\mathrm{d}_{s} A \wedge * \mathrm{~d}_{s} A\right)+A \wedge J \wedge \mathrm{~d} t \end{aligned}` | $\begin{aligned} \mathscr{L} & =-\frac{1}{2}\left( \mathrm{~d}_{t} A+\mathrm{d}_{s} A\right) \wedge *\left( \mathrm{~d}_{t} A+\mathrm{d}_{s} A\right)+A \wedge \mathscr{J} \\ & =-\frac{1}{2}\left(\mathrm{~d}_{t} A \wedge * \mathrm{~d}_{t} A+\mathrm{d}_{s} A \wedge * \mathrm{~d}_{s} A\right)+A \wedge J \wedge \mathrm{~d} t \end{aligned}$ | $\begin{aligned} \mathscr{L} &= -\frac{1}{2}(\mathrm{d}_t A + \mathrm{d}_s A) \wedge *(\mathrm{d}_t A + \mathrm{d}_s A) + A \wedge \mathscr{J} \\ &= -\frac{1}{2}(\mathrm{d}_t A \wedge * \mathrm{d}_t A + \mathrm{d}_s A \wedge * \mathrm{d}_s A) + A \wedge J \wedge \mathrm{d}t \end{aligned}$ |
| 0707.4470_EQ0061 | 32 | ■ 1.000 | `\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\int_{X} \mathrm{~d}_{s} f \wedge J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{s} J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{t} \rho=-\left.\int_{\Sigma} f \wedge \rho\right\|_{t_{0}} ^{t_{f}} .` | $\mathbf{d} S[A] \cdot \mathrm{d}_{s} f=\int_{X} \mathrm{~d}_{s} f \wedge J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{s} J \wedge \mathrm{~d} t=-\int_{X} f \wedge \mathrm{~d}_{t} \rho=-\left.\int_{\Sigma} f \wedge \rho\right\rvert_{t_{0}} ^{t_{f}} .$ | $\mathbf{d}S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d}t = -\int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d}t = -\int_X f \wedge \mathrm{d}_t \rho = -\int_\Sigma f \wedge \rho \Big\rvert_{t_0}^{t_f}.$ |
| 0707.4470_EQ0062 | 33 | ■ 1.000 | `\left.\left(\mathrm{d}_{s} D-\rho\right)\right\|_{t_{0}} ^{t_{f}}=0 .` | $\left.\left(\mathrm{d}_{s} D-\rho\right)\right\rvert_{t_{0}} ^{t_{f}}=0 .$ | $(\mathrm{d}_s D - \rho)\big\rvert_{t_0}^{t_f} = 0.$ |